

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	K. Abishek	Reg. No.:	192521120
Section / Batch:	IT	Date:	30/7/26

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I, K. Abishek / 192521120 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Abishek

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name: _____	Name: _____	Name: _____
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Binary Search - Search For Highest Element

Given, $A = [18, 16, 24, 32, 40, 48, 56, 64]$

Target - 64 $n = 8$

Initial:

Low = 0

High = 7

mid = $7/2 = 3$

$A[3] = 32$

Since $64 > 32$, Search right half

Remaining: $[40, 48, 56, 64]$

Comparisons = 1

Next:

Low = 4

High = 7

mid = 5

$A[5] = 48$

Since $64 > 48$, Search right half.

Remaining: $[56, 64]$

Comparisons = 2

Next:

Low = 6

High = 7

mid = 6

$A[6] = 56$

Since $64 > 56$, Search right half.

Remaining: $[64]$

Comparisons = 3

Next:

Low = 7

High = 7

mid = 7

$A[7] = 64$

Since $64 = 64$, element is Found.

Comparisons = 4

Search reduction: $8 \rightarrow 4 \rightarrow 2 \rightarrow 1$

Comparison Table:

Low	High	mid	$A[mid]$	Result
0	0	3	32	$64 > 32$
4	7	5	48	$64 > 48$
6	7	6	56	$64 > 56$
7	7	7	64	$64 = 64$

Total comparisons = 4

Time complexity:

$n \rightarrow n/2 \rightarrow n/4 \rightarrow n/8$

$T(n) = O(\log n)$

for $n = 8$

$\log_2 8 = 3$

space complexity:

only Low, High, mid variables used.

space complexity = $O(1)$

Efficiency:

Binary Search moves right half at each Step.

$\therefore$ Thus, only 4 comparisons are required instead of checking all 8 elements sequentially.

Conclusion:

Binary Search is efficient for the given dataset since it reduces the Search Space by half at every comparison.

The element 64 is found at index 7 in 4 comparisons with a time complexity of $O(\log n)$.